Adaptive Lane Detection Under Varying Lighting for Resource-Constrained Autonomous Systems

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Abstract. Robust lane detection under varying illumination remains a major challenge for autonomous vehicles and Advanced Driver Assistance Systems (ADAS), particularly on resource-constrained embedded platforms where deep learning methods are often impractical due to their computational requirements. Classical lane detection methods are computationally efficient but often fail under shadows and night-time conditions. This paper proposes an adaptive lane detection method that classifies each frame according to its illumination condition and applies a dedicated processing pipeline with continuously adjusted HSV thresholding and specialised night-time enhancement. A two-zone lane position estimation strategy is further introduced to improve detection stability. The proposed method is evaluated on the CULane benchmark against Hough Transform and fixed HSV baseline methods. Experimental results demonstrate that the proposed approach maintains substantially more consistent performance under challenging illumination conditions, particularly under shadow and night-time scenarios, while maintaining real-time performance on CPU-only hardware. The lightweight and training-free nature of the proposed method makes it suitable for embedded autonomous driving applications.

Keywords: Lane detection · Adaptive thresholding · Illumination robustness · CPU-only deployment · CULane benchmark · Classical computer vision

1 Introduction

Robust lane detection under varying illumination remains one of the most critical and challenging perception tasks for autonomous vehicles, directly impacting vehicle safety and path planning reliability. Cameras are among the most practical and cost-effective sensors and lane detection is a fundamental function of these vision-based systems that provides the critical information for lane keeping, vehicle positioning, and path planning [1]. So, developing lane detection methods that are both accurate and capable of operating in real time on resource-constrained hardware remains an important research challenge.

Despite significant progress, lane detection under changing illumination conditions presents several fundamental challenges. Under night-time conditions,

lane markings exhibit significantly reduced contrast against the road surface, making them difficult to distinguish using fixed colour or edge thresholds. Shadows cast by roadside structures or vehicles create strong edges that are easily misclassified as lane boundaries by edge-based detectors.

Classical lane detection methods commonly rely on colour thresholding and edge-based techniques such as the Hough Transform [2]. While computationally efficient, their performance degrades significantly under shadows and night-time conditions due to their reliance on fixed parameters. Deep learning methods, including SCNN [10], RESA [24], and end-to-end architectures [16], achieve higher detection accuracy but typically require large labelled datasets and GPU acceleration, limiting their deployment on embedded platforms.

Several studies have attempted to improve classical lane detection under varying illumination. Mohamed Abdelaal et al. [8] proposed a system that switches between separate daytime and night-time pipelines based on scene brightness. However, discrete mode switching can become unstable under transitional lighting conditions such as dawn, dusk, or tunnel entry/exit, where scene brightness fluctuates rapidly and abrupt pipeline switching leads to inconsistent detection. Other approaches incorporate tracking mechanisms [3] or geometric transformations [4], but their pipelines remain largely fixed after configuration.

To address these limitations, this paper proposes an adaptive lane detection framework that continuously adjusts its processing strategy according to the brightness of each frame. The method employs illumination-aware frame classification, adaptive HSV thresholding that modifies colour thresholds under different lighting conditions, and a dedicated night-time enhancement pipeline for robust low-light detection. A two-zone weighted centroid method with dead-zone stabilisation and lane-lost recovery provides stable lane tracking. The proposed method is designed for real-time operation on resource-constrained CPU-only hardware without requiring GPU acceleration. The main contributions of this paper are summarised as follows:

- We propose an adaptive lane detection framework that classifies frames into day, shadow, and night conditions and applies a dedicated processing pipeline for each scenario, enabling continuous adaptation to changing illumination without discrete mode-switching artefacts.
- We introduce a continuous adaptive HSV thresholding strategy that adjusts colour thresholds on a per-frame basis, together with a night-time enhancement pipeline, to improve robustness under varying illumination.
- We develop a two-zone weighted centroid lane position estimation method with dead-zone stabilisation and lane-lost recovery with timeout, ensuring stable lane tracking during transient detection failures.
- We evaluate the proposed method on the CULane benchmark against Hough Transform and fixed HSV baseline methods, demonstrating improved robustness under challenging illumination conditions while maintaining real-time performance on CPU-only hardware.

2 Related Work

2.1 Classical Lane Detection

Classical lane detection methods rely on hand-crafted features and geometric models, offering low computational overhead suitable for resource-constrained hardware. The Hough Transform [2] is one of the most widely used techniques for extracting lane lines. To improve efficiency, the Probabilistic Hough Transform [5] and Progressive Probabilistic Hough Transform [6] reduce the number of processed edge points and are commonly combined with Canny edge detection [7]. However, these methods remain highly sensitive to illumination changes, with performance degrading significantly under shadows, glare, and night-time conditions due to weak or fragmented edge responses.

To improve robustness, Abdelaal et al. [8] proposed an adaptive lane detection system that switches between daytime and night-time processing pipelines based on scene brightness, combining Hough Transform-based lane detection with a Kalman filter. However, the use of fixed brightness thresholds can lead to unstable performance under conditions, such as dawn, dusk, and tunnel entry or exit.

2.2 Adaptive and Tracking-Based Classical Methods

To overcome the limitations of fixed-threshold edge detection, several classical approaches have incorporated tracking and adaptive mechanisms to improve robustness under challenging conditions. Sultana and Ahmed [3] proposed a real-time night-time lane detection method to improve reliability under low-light conditions, where conventional daytime techniques often fail. Aly [4] addressed urban lane detection using Inverse Perspective Mapping, Gaussian filtering, Hough-based line detection, and RANSAC fitting to handle both straight and curved lanes. Son et al. [17] developed an illumination-invariant lane detection method for lane departure warning systems, improving robustness under varying lighting conditions. More recently, Hasan et al. [11] proposed a training-free classical lane detection approach for day-to-night operation on CPU-only platforms, combining contrast enhancement, adaptive edge detection, trapezoidal ROI masking, and Hough Transform-based lane estimation. However, the method employs a fixed region of interest and lacks continuous adaptation to rapidly changing illumination conditions.

2.3 Deep Learning-Based Lane Detection

Early architectures such as SCNN [10] and RESA [24] incorporated structural priors to capture lane continuity, while LaneATT [12] employed an anchor-based attention mechanism to balance accuracy and real-time performance. More recently, Transformer-based approaches have gained prominence. Lane2Seq [19] reformulates lane detection as a sequence generation problem, whereas LDTR [20] adopts an anchor-chain representation to improve robustness under occlusion.

For curved roads, BSNet [21] uses B-spline representations to achieve efficient curve-based detection, while LaneTCA [22] incorporates temporal attention to improve prediction stability across video frames.

Li and Hou [18] combined adaptive image enhancement with wavelet-based feature extraction for shadow and night-time detection, while Peng et al. [23] introduced the NightLane dataset and proposed the FLLENet low-light enhancement framework. Despite their high accuracy, deep learning methods typically require large labelled datasets, computationally intensive architectures, and GPU acceleration, limiting their deployment on resource-constrained embedded platforms.

Beyond computational requirements, system certification presents additional challenges. ISO 26262 [15] defines strict functional safety requirements, where the deterministic behaviour and interpretability of classical computer vision methods offer advantages over black-box neural networks. As highlighted by recent surveys [14], classical approaches also maintain lower computational and memory requirements, motivating the continued development of lightweight lane detection methods for embedded systems operating under varying illumination conditions.

3 Proposed Method

3.1 Overview of the Pipeline

This study proposes an adaptive lane detection system with dedicated processing pipelines for daytime, shadow, and night-time conditions. For daytime and shadow scenes, the system uses adaptive HSV thresholding and morphological filtering, while night-time scenes employ additional enhancement techniques, including Contrast Limited Adaptive Histogram Equalization (CLAHE) and vertical top-hat filtering, to improve low-light detection. The appropriate pipeline is selected automatically through real-time brightness analysis, enabling continuous adaptation to changing illumination conditions. After lane mask generation, a two-zone mode-centroid method estimates the lane position, while dead-zone stabilisation and a lane-lost fallback mechanism improve detection stability. Fig 1 illustrates the overall architecture of the proposed system.

3.2 Pre-processing and Colour Space Conversion

Each input frame is first resized to 160×120 pixels to reduce computational cost while preserving sufficient details for lane detection. This enables real-time processing on resource-constrained hardware. A 5×5 Gaussian blur is then applied to reduce noise and smooth the image. Finally, the frame is converted from BGR to HSV colour space using OpenCV's `cv2.cvtColor()` function. The HSV representation separates colour and brightness information, allowing more effective adaptive thresholding under varying illumination conditions.

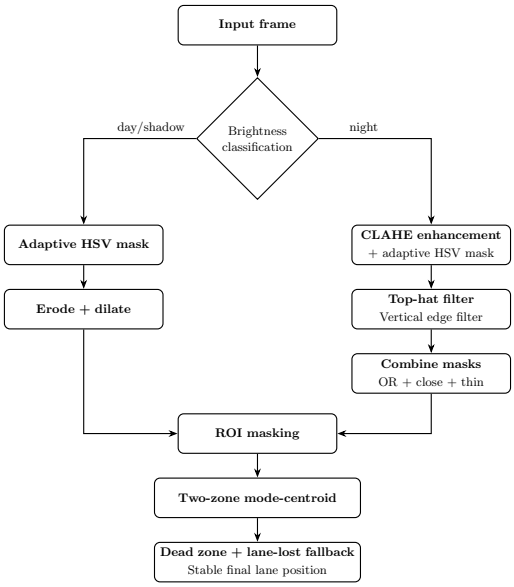


Fig. 1: Overview of the proposed adaptive HSV two-zone lane detection.

3.3 Lighting Condition Classification

To adapt to varying illumination conditions, each frame is first classified according to its lighting level. The input frame is converted to grayscale, and its mean pixel intensity is computed to estimate the overall scene brightness. Based on this value, frames are classified as night(brightness < 60), shadow ($60 \leq$ brightness < 100), or day (brightness ≥ 100). These thresholds were determined empirically through iterative testing on the CULane dataset: values below 60 consistently correspond to night-time conditions where contrast is severely reduced; values between 60 and 100 correspond to shadowed regions; values above 100 correspond to well-lit daytime scenes. This classification enables the system to apply the appropriate processing pipeline for each condition. Night-time frames are processed using the enhanced pipeline described in section 3.4, while day and shadow frames follow the standard pipeline described in section 3.5.

3.4 Night-Time Processing Pipeline

Frames classified as night undergo a multi-stage processing pipeline designed to improve lane visibility under low-light conditions.

Contrast Enhancement. The input frame is converted to LAB colour space, and CLAHE is applied to the luminance channel using a clip limit of 3.0 and a tile grid size of 8×8 to enhance local contrast and improve lane visibility under low-light conditions. The enhanced image is then converted back to BGR colour space.

Adaptive HSV Thresholding. The enhanced frame is converted to HSV colour space, and the average brightness of the Value channel, V_{mean}, is computed. The adaptive thresholds are defined as:

$$\text{lower_V} = 10 + \frac{V_{\text{mean}}}{255} \times 50 \tag{1}$$

$$\text{upper_S} = 100 - \frac{V_{\text{mean}}}{255} \times 30 \tag{2}$$

The lower Value threshold typically varies between 10 and 60, while the upper Saturation threshold ranges from 100 to 70. For robustness, the lower Value threshold is clamped to the range [10, 80]. The remaining HSV parameters are fixed, and a binary lane mask is generated using OpenCV's `cv2.inRange()` function.

Vertical Top-Hat Transform. A morphological Top-Hat transform is applied to the gamma-corrected image using a 3×21 vertical structuring element:

$$I_{\text{TopHat}} = I - (I \circ S) \tag{3}$$

where S is the structuring element and $\circ$ denotes the morphological opening operation. This operation enhances vertical lane structures while suppressing background noise. The resulting image is thresholded at an intensity of 10 to produce a secondary binary mask.

Mask Fusion. The HSV and Top-Hat masks are combined using a logical OR operation:

$$M_{\text{fused}} = M_{\text{HSV}} \vee M_{\text{TopHat}} \tag{4}$$

This fusion allows both colour and structural information to contribute to lane detection.

Morphological Cleaning. The fused mask is refined using morphological closing with a 3×3 kernel, followed by dilation with a 3×3 kernel and erosion with a 2×2 kernel, to improve lane continuity while suppressing noise.

Tiny Component Removal. Finally, connected components with an area smaller than 10 pixels are removed to eliminate isolated noise and spurious detections before lane position estimation.

3.5 Day and Shadow Processing Pipeline

For day and shadow conditions, where lane markings remain relatively visible, a simplified processing pipeline is used. The input frame is converted to HSV colour space, and adaptive thresholding is applied using tighter thresholds than those used for night-time processing:

$$\text{lower_V} = 20 + \frac{V_{\text{mean}}}{255} \times 100 \tag{5}$$

$$\text{upper_S} = 80 - \frac{V_{\text{mean}}}{255} \times 30 \tag{6}$$

The remaining HSV parameters are fixed, and a binary lane mask is generated using OpenCV's `cv2.inRange()` function. The mask is then refined through morphological erosion and dilation to remove noise and improve lane continuity before lane position estimation.

3.6 Region of Interest Masking

A trapezoidal region of interest (ROI) is applied to both the night and day/shadow lane masks to restrict processing to the drivable area. For a frame with width W and height H, the ROI vertices are defined as:

$$\begin{aligned} \text{Top-left} &= (0.20W,\ 0.45H), \qquad \text{Top-right} = (0.80W,\ 0.45H) \\ \text{Bottom-left} &= (0,\ H), \qquad \text{Bottom-right} = (W,\ H) \end{aligned} \tag{7}$$

The trapezoidal ROI approximates the perspective view of a forward-facing camera, where lane markings converge toward the horizon. This masking step removes irrelevant regions, such as the sky and roadside areas, ensuring that subsequent processing focuses only on the road surface.

3.7 Two-Zone Weighted Lane Position Estimation

After ROI masking, the lane position is estimated using a two-zone weighted approach. The masked region is divided into a near zone (lower half) and a far zone (upper half). For each zone, a histogram of lane pixel positions is computed, and the histogram peak is selected as the representative lane position. This approach is more robust to noise and spurious detections than a simple mean. If fewer than five lane pixels are detected in a zone, the frame centre is used as the default position.

The final lane position is calculated as:

$$x_{\text{final}} = 0.70\, c_{\text{near}} + 0.30\, c_{\text{far}} \tag{8}$$

where c_{near} and c_{far} are the lane positions estimated from the near and far zones, respectively. The near zone is assigned a higher weight to prioritise immediate steering information, while the far zone provides look-ahead information for upcoming road curvature.

3.8 Dead Zone Stabilisation and Lane-Lost Fallback

To reduce unnecessary steering corrections caused by small fluctuations in the estimated lane position, a dead zone is defined around the frame centre. If the estimated lane position, x_{final}, lies within 12% of the frame width from the centre, it is snapped to the frame centre, $x_c = W/2$,

$$x_{\text{final}} = \begin{cases} x_c, & \text{if } |x_{\text{final}} - x_c| < 0.12W \\ x_{\text{final}}, & \text{otherwise} \end{cases} \tag{9}$$

This stabilisation reduces minor positional fluctuations and produces smoother steering output. To improve robustness during temporary detection failures, the proportion of detected lane pixels within the ROI is monitored continuously. If this proportion falls below 0.5%, the previous lane position is reused as a fallback estimate, preventing unstable lane predictions during brief periods of poor visibility.

4 Experiments

4.1 Datasets

We evaluate our proposed method on the **CULane** [10] benchmark dataset.

CULane [10] is a large-scale lane detection dataset containing 133,235 images collected from real urban and highway driving scenarios. In this study, the 34,680-image test set is used for evaluation. The dataset includes diverse lighting conditions, including normal, shadow, dazzle, and night scenes, making it well suited for assessing the robustness of the proposed adaptive thresholding method. The results are reported separately for these four subsets to evaluate performance under different illumination conditions.

4.2 Baseline Methods

To evaluate the proposed adaptive lane detection method, two classical computer vision approaches are used as baselines. The first baseline is a conventional edge-based lane detector, while the second is a fixed HSV thresholding method. These baselines provide a reference for assessing the benefits of adaptive thresholding and night-time enhancement under varying illumination conditions.

4.3 Method 1 — Hough Transform

The first baseline employs a classical edge-based lane detection pipeline. Each frame undergoes grayscale conversion, contrast enhancement, Gaussian smoothing, and Canny edge detection, followed by the Probabilistic Hough Transform within the same trapezoidal ROI used by the proposed method. The detected line segments are rendered onto a binary mask to produce the final lane output. As this method relies solely on edge information without adaptive thresholding, it serves as a representative classical baseline for comparison.

4.4 Method 2 — Fixed HSV Two-Zone Weighted Centroid

The second baseline employs a fixed-threshold colour segmentation approach. Each frame is converted to HSV colour space, where fixed threshold ranges are used to detect lane markings. Unlike the proposed method, these thresholds remain constant and are not adapted to scene brightness. The resulting binary mask is refined through morphological operations, after which the same two-zone weighted centroid method used in the proposed approach is applied to estimate the final lane position. This baseline is included to evaluate the effectiveness of adaptive thresholding under varying illumination conditions.

4.5 Evaluation Metrics

The following evaluation metrics are used to assess the performance of all the methods.

Accuracy. Accuracy measures the percentage of correctly detected ground-truth lane points within a 10-pixel tolerance:

$$\text{Accuracy} = \frac{TP}{TP + FN} \times 100 \tag{10}$$

where TP and FN are computed by comparing the predicted lane mask with the rendered ground-truth mask.

Precision and Recall. Precision and recall are computed at the pixel level as:

$$\text{Precision} = \frac{TP_{\text{pixels}}}{TP_{\text{pixels}} + FP_{\text{pixels}}} \times 100 \tag{11}$$

where FP denotes false-positive lane pixels.

F1-Score. The F1-score is calculated as the harmonic mean of precision and recall:

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \times 100 \tag{12}$$

Intersection over Union (IoU). IoU measures the overlap between the predicted and ground-truth lane masks and is defined as:

$$\text{IoU} = \frac{|\text{Prediction} \cap \text{Ground Truth}|}{|\text{Prediction} \cup \text{Ground Truth}|} \times 100 \tag{13}$$

We report the average IoU over all evaluated frames as a measure of mask overlap quality under different illumination conditions.

Runtime Performance. Real-time performance is evaluated using average latency and frames per second (FPS):

$$\text{Latency (ms)} = \frac{1}{N}\sum_{i=1}^{N}(T_{\text{end},i} - T_{\text{start},i}) \times 1000 \tag{14}$$

$$\text{FPS} = \frac{1000}{\text{Latency (ms)}} \tag{15}$$

where timing measurements include only the lane detection pipeline.

Lane-Lost Rate. Robustness during detection failures is evaluated using the lane-lost rate:

$$\text{Lane-Lost (\%)} = \frac{\text{Frames with foreground pixel proportion} < 0.005}{N} \times 100 \tag{16}$$

where the foreground ratio is computed within the region of interest.

Illumination-Specific Evaluation. To assess robustness under varying lighting conditions, frames are classified according to their mean grayscale intensity as night (< 60), shadow ($60 \leq I < 100$), and day (≥ 100), following the thresholds defined in section 3.3.

4.6 Implementation Details

All methods were implemented in Python using OpenCV and NumPy, without the use of deep learning frameworks or GPU acceleration. Experiments were conducted on an 8th Gen Intel Core i7 CPU with 8 GB RAM.

All input frames were resized to 160×120 pixels, reducing the pixel count by approximately 97% compared to the original CULane resolution (1640×590). Frames were processed sequentially without batching or parallelisation.

A common evaluation pipeline with a 10-pixel matching tolerance was used for all methods, and ground-truth annotations were scaled to match the processing resolution.

Runtime performance was measured using Python's `time.perf_counter()`, with metric computation excluded. Timing measurements included only the lane detection pipeline, from pre-processing to final lane output.

4.7 Results

Quantitative Results. Table. 1 presents the overall performance of the three methods on the CULane test set, while Table. 2 reports accuracy under different illumination conditions. The proposed method achieves the highest overall accuracy (71.7%) and IoU (8.8%), while maintaining an F1-measure comparable to the Hough baseline.

Table 1: Overall performance comparison on CULane.

Method	Accuracy (%)	F1 (%)	IoU (%)
Method 1 (Hough)	68.8	10.0	8.6
Method 2 (Fixed HSV)	55.4	9.7	8.0
Method 3 (Proposed)	**71.7**	10.0	**8.8**

Table 2: Accuracy by lighting condition on CULane.

Method	Day (%)	Shadow (%)	Night (%)
Method 1 (Hough)	**77.9**	**76.1**	54.1
Method 2 (Fixed HSV)	72.7	20.9	36.9
Method 3 (Proposed)	71.2	73.4	**70.8**

The results indicate that the performance improvement is primarily driven by challenging illumination conditions. Under daytime conditions, the Hough baseline achieves the highest accuracy, with the proposed method performing comparably. However, under shadow conditions, the fixed HSV baseline degrades significantly to 20.9%, reflecting its inability to adapt to reduced contrast using a fixed Value threshold. Under night-time conditions, the proposed method achieves 70.8% accuracy, substantially outperforming both the Hough baseline (54.1%) and the fixed HSV baseline (36.9%). These results demonstrate the effectiveness of the proposed adaptive thresholding and night-time enhancement pipeline. Overall, although the proposed method does not achieve the best performance in every scenario, it provides the most consistent performance across varying illumination conditions.

Qualitative Results. Figures 2, 3, and 4 represent the detection results from all three methods under daytime, shadow, and night-time conditions respectively. Under daytime conditions, all three methods produce acceptable lane detections, consistent with their comparable quantitative performance. Under shadow conditions, the fixed HSV baseline generates incomplete lane masks due to its inability to adapt to reduced contrast, whereas the proposed method maintains more continuous lane detection through adaptive thresholding. Under night-time conditions, the Hough baseline frequently fails because of insufficient edge contrast, while the fixed HSV baseline produces sparse and unreliable detections. In contrast, the proposed method maintains robust lane detection through its dedicated night-time enhancement pipeline, which combines CLAHE, adaptive HSV thresholding, and Top-Hat filtering.

Ablation Study To evaluate the contribution of each component in the proposed pipeline, an ablation study is conducted on the CULane night subset. Table. 3 reports the performance of the full method and variants with individual components removed.

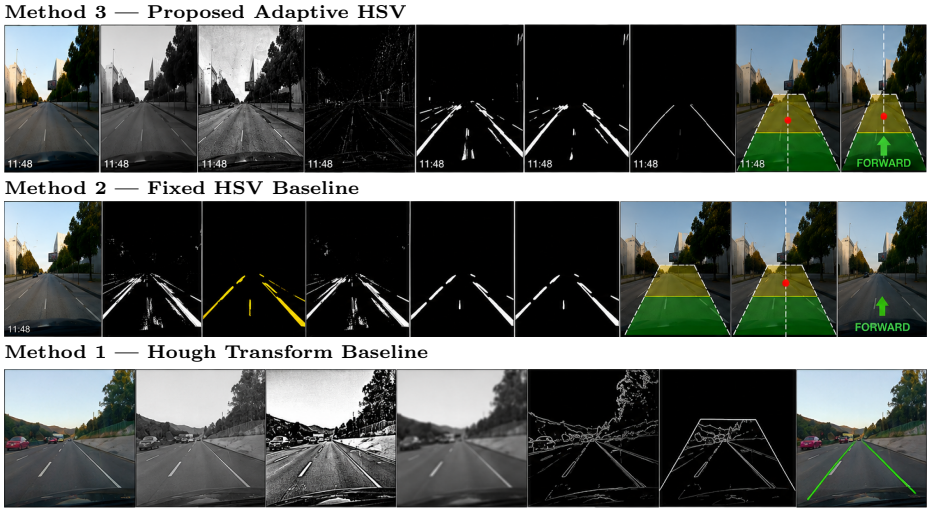


Fig. 2: Comparison of lane detection under daytime conditions on CULane. Top row: Method 3 (Proposed Adaptive HSV), showing intermediate pipeline stages from input frame to final lane. Middle row: Method 2 (Fixed HSV Baseline). Bottom row: Method 1 (Hough Transform Baseline). The *green line* indicates the detected lane position, and the *red dashed line* shows the zone boundary.

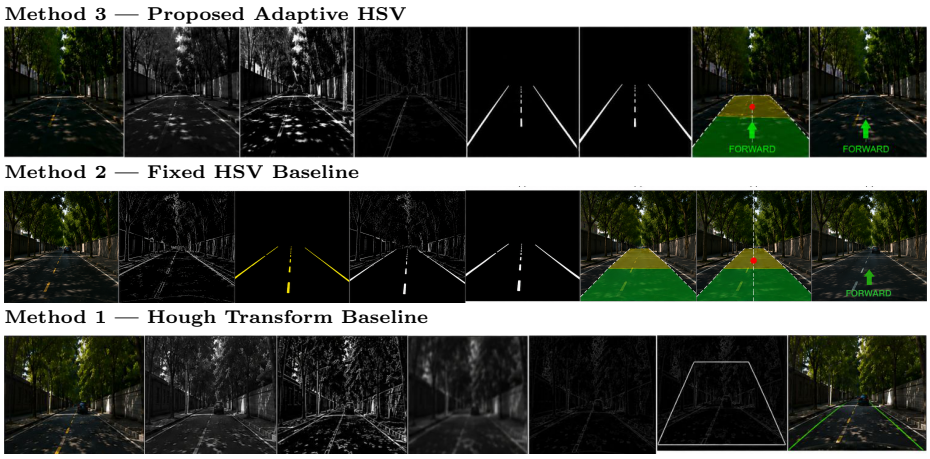


Fig. 3: Comparison of lane detection under shadow conditions on CULane. Top row: Method 3 (Proposed Adaptive HSV), showing intermediate pipeline stages from input frame to final lane. Middle row: Method 2 (Fixed HSV Baseline). Bottom row: Method 1 (Hough Transform Baseline). The *green line* indicates the detected lane position, and the *red dashed line* shows the zone boundary.

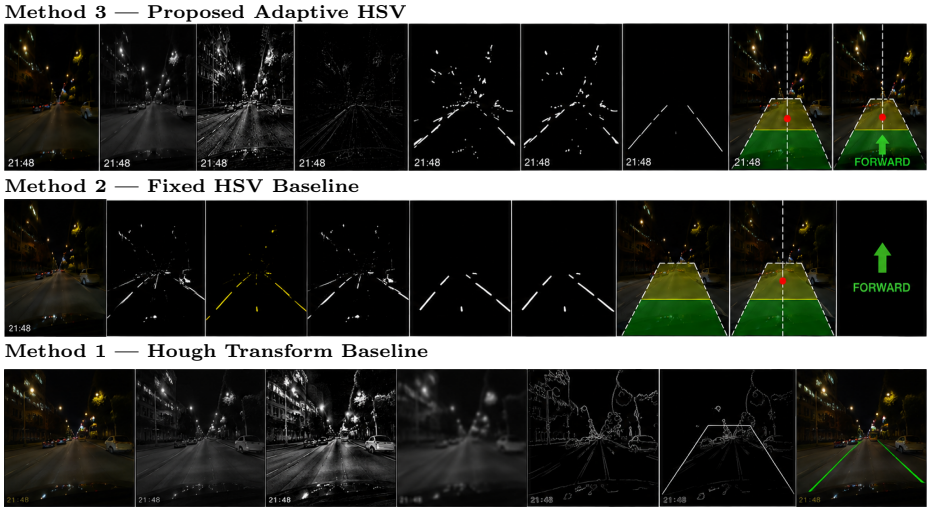


Fig. 4: Comparison of lane detection under night-time conditions on CULane. Top row: Method 3 (Proposed Adaptive HSV), showing intermediate pipeline stages from input frame to final lane. Middle row: Method 2 (Fixed HSV Baseline). Bottom row: Method 1 (Hough Transform Baseline). The *green line* indicates the detected lane position, and the *red dashed line* shows the zone boundary.

Table 3: Ablation study on the CULane night subset.

Variant	Overall Acc (%)	Night Acc (%)	Night F1 (%)
Full (Proposed)	**71.7**	**70.8**	9.6
w/o CLAHE	64.7	51.5	11.4
w/o Top-Hat	66.5	56.6	8.5
w/o Adaptive HSV	43.1	44.3	12.1
w/o Two-Zone	71.7	70.8	9.6

Table 4: Runtime performance on Intel Core i7-1135G7 CPU.

Method	Time per Frame (ms)	FPS
Method 1 (Hough)	2.1	466.4
Method 2 (Fixed HSV)	0.9	1156.3
Method 3 (Proposed)	1.5	686.8

The results show that adaptive HSV thresholding provides the largest contribution to performance, with its removal reducing overall accuracy by 28.6 percentage points. Removing CLAHE and Top-Hat filtering decreases night-time accuracy by 19.3 and 14.2 percentage points, respectively, confirming the importance of contrast enhancement and structural feature extraction under low-light conditions. In contrast, removing the two-zone weighting strategy has minimal impact on the evaluated metrics, indicating that its primary benefit lies in improving lane position stability rather than detection accuracy. Overall, the ablation study demonstrates that the combination of adaptive thresholding, contrast enhancement, and structural filtering is essential for robust lane detection under challenging illumination conditions.

Runtime Performance. Table. 4 reports the average processing time and frames per second (FPS) for each method on the target CPU platform. All three methods operate well above the real-time requirements of ADAS applications. The fixed HSV baseline achieves the highest processing speed due to its simple thresholding pipeline. Despite incorporating additional enhancement and filtering stages for night-time processing, the proposed method maintains real-time performance with only modest computational overhead, demonstrating its suitability for deployment on resource-constrained embedded platforms.

5 Conclusion

In this paper, we proposed an adaptive lane detection method for resource-constrained autonomous systems operating under varying illumination conditions. The proposed approach combines brightness-based scene classification, adaptive HSV thresholding, and dedicated night-time enhancement techniques to improve lane detection under day, shadow, and night conditions. A two-zone lane position estimation strategy, together with stabilisation and lane-lost recovery mechanisms, further improves detection reliability.

The proposed method was evaluated on the CULane dataset and compared with Hough Transform and fixed HSV baseline methods. Experimental results demonstrate that the proposed approach provides more consistent performance under challenging illumination conditions, particularly in shadow and night-time scenarios. Ablation studies further confirm the importance of adaptive thresholding and night-time enhancement.

Overall, the results show that classical computer vision methods can remain effective when designed to adapt to changing environmental conditions. The lightweight and training-free nature of the proposed method makes it suitable for embedded and resource-constrained autonomous driving applications.

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